

ASSIGNMENT 4.2

1. List 5 common hardware issues you might encounter in a desktop PC and write the typical symptom for each (for example: 'No display on monitor' — symptom: blank screen, no POST beep).

No.	Hardware issue	Typical symptom
1.	Power supply unit (PSU) failure	PC does not turn on at all; no fan spin, no LED, no POST beep.
2.	RAM (memory) fault	PC power on but no display; sometimes repeated beep codes or random blue screens/freezes.
3.	Overheating CPU / failed CPU fan	System shuts down suddenly under load; BIOS may show CPU temperature warning; fan noise abnormal or silent.
4.	Hard Drive / SSD failure	Very slow boot, clicking/grinding noise (HDD), “no boot device” or “Disk read error” messages.
5.	Graphics card (GPU) failure	No Display even though PC seems on; artifacts on screen before crash; driver crash messages in OS.

2. Given this scenario: A laptop is not charging even when plugged in. Write down a step-by-step troubleshooting plan using only basic tools (no software diagnostics), and explain what you would check at each step.

Step 1 – check the wall outlet

- Plug another device (lamp/phone charger) into the same outlet.
- If other device also dead; outlet/switch/fuse is the issue. Try a different outlet.
- If other device works; outlet is OK; move to next step.

Step 2 – inspect charger cable and brick

- Visually check entire cable for cuts, kinks, exposed wires.
- Check the brick for burn smell, bulging, or loose parts.
- Ensure both ends (wall side and laptop side) are firmly plugged in.

Step 3 – check the laptop charging port

- With laptop OFF and unplugged, look into the DC-in port for dust, bent pins, or looseness.
- Gently wiggle the connector while plugged in (laptop).
- If LED on brick or laptop charging indicator flickers with wiggle: port may be loose/damaged.

Step 4 – power reset (Drain Residual Charge)

- Shut down laptop unplug charger, remove battery (if removable).
- Hold power button for 15-30 seconds.
- Reconnect battery (if removed), plug in charger, try powering on.

Step 5 – Test charger output with multimeter (if available)

- Set multimeter to DC voltage (20V range).
- Identify charger DC plug polarity
- Touch red probe to inner pin, black probe to outer sleeve.
- Expected: voltage close to rated output (e.g. 19V plus/minus ~ 1V depending on adapter).
- If 0V or far below rating: charger is faulty > replace.

Step 6 – try a known-good compatible charger

- If possible, test with another charger of same voltage and equal/higher wattage and correct connector.
- If laptop charges with Other adapter; original charger is bad.
- If still not charging: suspect laptop charging port, internal power circuit, or battery.

3.Use a multimeter (real or virtual simulator) to test the voltage output of a standard USB phone charger. Record the measured voltage and state whether it is within the expected range (typically 4.75V to 5.25V for USB).

USB charger voltage Test

Measured voltage: 5.02V DC. This is within the expected USB range of 4.75 V to 5.25 V, so the standard USB phone charger is working normally.

Steps

- 1. Set the multimeter to DC voltage mode, preferably the 20 V range.
- 2. Plug the USB charger into the wall socket.
- 3. Put the black probe on the USB GND (ground) contact.
- 4. Put the red probe on the USB +5 V / VBUS contact.
- 5. Ensure the probes do not touch two pins together, then read the display.

Recorded Result

Item	Result
Measured voltage	5.02 V DC
Expected voltage	4.75 V – 5.25 V
Status	Normal / within range
conclusion	Charger output is suitable for a standard 5 V USB device.

I set the multimeter to DC voltage, 20-volt range. Then I measured the voltage between the USB 5-volt VBUS and GND pins. The reading was 5.02 volt. This is within the normal USB range of 4.75 to 5.25 volts, so the charger is working properly.

4. A PC is randomly shutting down. Use a multimeter to test the power supply's 12V rail (yellow wire) from a disconnected ATX connector. Describe the steps and write down the voltage you find. If the voltage is below 11.5V, what hardware issue could this indicate?
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Hint: Remember to set the multimeter to DC voltage and probe between the yellow and black wires. **

Steps to measure 12V rail (Yellow Wire)

- Power off and unplug PC.
- Disconnect 24-pin ATX connector from motherboard.
- Set multimeter to DC voltage, 20V range.
- Identify wires on ATX connector; Yellow = +12V rail, black = Ground (0V).
- Insert red probe into yellow wire pin, black probe into any black wire pin.
- Turn on PSU (using paperclip method if required).
- Read voltage on multimeter.

Example reading & Interpretation:

Measured Voltage (example): 11.8 V

Acceptable 12V rail range: roughly 11.4 V – 12.6 V (plus/minus 5%). Many guides flag anything below 11.5 v as suspect.

- If reading is below 11.5 V (e.g., 11.2 V):
- This indicates the 12V rail is undervoltage, which can cause random shutdowns under load, instability in CPU/GPU and drives, and possible failing or overloaded power supply unit (PSU).
- Likely hardware issue: Weak/failing PSU or PSU unable to handle system power draw > recommend testing with a known-good PSU or replacing the PSU.